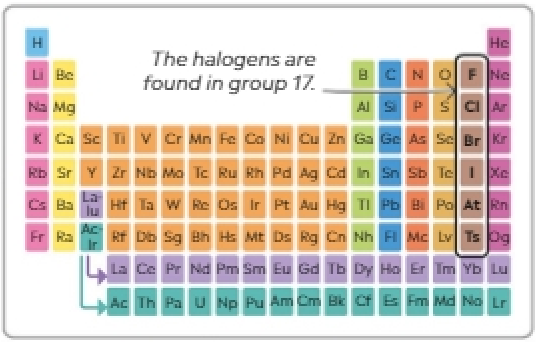

Halogens

The word halogen means "salt forming." Elements in this group form salts when they react with metals. For example, lithium bromide is a compound of the halogen bromine and the metal lithium, while table salt—sodium chloride—is a compound of chlorine and sodium. The first four members of this group—fluorine, chlorine, bromine, and iodine—are all very reactive in nature. Astatine is far less reactive, while the chemistry of tennessine—a halogen confirmed in 2016—is still a mystery.

Chlorine

17 Cl

This halogen and its compounds are both dangerous and useful. Pure chlorine gas is toxic and has been used as a weapon, but chemicals known as hypochlorite salts are very effective at killing bacteria in water—making it safe to swim in. Chlorine dioxide is added to water to make sure it is safe to drink.

Common salt

Before it is purified and placed at the dinner table, table salt can be found as a crystalline mineral in its natural form. Known commonly as rock salt, its scientific name is sodium chloride.

This tube shows the level of chlorine in the water. The darker the sample, the more chlorine in the pool.

This tube indicates the pH of the water. The more yellow the sample, the lower the pH value, and the greater the acidity.

Halite

Chlorine can be extracted from the mineral halite using a process called electrolysis. Dissolving halite in water and passing electricity through it releases chlorine gas. Huge amounts of chlorine can also be found in the ocean.

Purifying water

Chlorine is used to disinfect water in swimming pools. It helps kill bacteria. It's important that the water's acidity is monitored so that the chemicals work effectively. This is measured using the pH scale, which ranges from 0 to 14. A pH of about 7.4–7.6 is ideal.